

Supplementary Information

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This file contains Supplementary Notes 1–9, legends for Supplementary Tables 1–67 (provided as Supplementary_Tables.xlsx) and legends for Supplementary Data 1–3 (provided as separate Excel files).

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Supplementary Note 1. Evidence archive and scope

The analysis is retrospective and read-only. It uses 4,240 archived runs: BixBench-Verified-50 (50 tasks; 1,500 runs, five configurations × two environments × three replicates), CompBioBench (100 tasks; 2,400 paired runs from four configurations plus 100 code-only GPT-6 Astra runs) and IWC (10 tasks; 240 runs, four configurations). Of these, 4,228 retain a primary agent event log (Codex JSONL, or Claude Code stream-JSON for the superseded DeepSeek harness); the 12 missing traces are CompBioBench runs. Galaxy job records comprise 23,080 non-fetch jobs (5,042 BixBench, 16,686 CompBioBench, 1,352 IWC), deduplicated by server and job identifier; data-fetch jobs are excluded. Detailed history snapshots exist for 714/750, 1,198/1,200 and 118/120 Galaxy runs; they were taken after the runs completed.

Archive integrity was checked before analysis: 160 source-evidence hashes were verified, all 4,240 run identifiers are unique, and the archived accuracy, job-state and token medians were reproduced from the raw records. No score was regraded; the original evaluator outcome of every run is retained, including scored missing answers. No agent code was executed and no Galaxy server was contacted. BixBench reference values were read only from the evaluator records of completed runs.

The three benchmarks retain different endpoints — binary evaluator acceptance (BixBench), archived aggregate score vectors without item-level grades (CompBioBench) and continuous 0–1 agreement with workflow reference outputs (IWC) — and are analysed as separate strata throughout. Extended Data Fig. 1a summarizes the archive and Extended Data Fig. 1b the audit pipeline.

Supplementary Note 2. Trace extraction, call classes and interface bypass

Every primary event log was parsed. For each Galaxy MCP call we recovered the tool name, arguments, structured status, error text, failure summary, validation errors and parameter provenance. Calls were grouped as *discovery* (search_galaxy_tools, inspect_galaxy_tool), *history inspection* (history and archive inspection), *execution* (run_galaxy_tool_and_wait, run_galaxy_udt_and_wait, job waits) and shell commands. Token totals are recorded per run and cannot be attributed to individual calls, so the number of characters each call returned to the agent is used as a proxy for context consumed (Fig. 5b).

Shell commands in Galaxy-environment runs were counted as Galaxy API calls when they used BioBlend or GalaxyInstance or called the Galaxy histories, datasets, jobs or tools API. Commands referencing BioBlend, GalaxyInstance or any /api/ path were matched against five operations: seed-history copy (copy_history, /copy), dataset download (download_dataset, /display, /download), tool-schema fetch (show_tool, /build), job polling (show_job, /jobs/{id}, wait_for_job) and raw job submission (run_tool, or a POST to /api/tools through requests, curl or urllib). Bypass, substitution and UDT statistics use Codex-harness traces (1,908 Galaxy runs: 600 BixBench, 1,188 CompBioBench, 120 IWC), because the Claude Code harness renders tool results in a different structure. The seed history is provided, and its copy required by the harness policy, only in BixBench.

Supplementary Note 3. Classification of failed Galaxy MCP calls

Of 69,812 Galaxy MCP calls, 7,389 returned a failure status. Each failed call was assigned one class by applying ordered regular-expression rules to its error text and status (first match wins; rules in Supplementary Table 34). Classes A1–A8 are failures before any job existed (3,454 calls); B1–B5 are failures of a created job (3,677); 258 calls matched no rule and are reported as unclassified. Rule order resolves overlaps; for example, a missing history context is assigned A1 even when the message also contains a validation keyword. Classes were designed to be actionable: each maps onto an interface change (Extended Data Fig. 3d).

Class	Name	Definition
A1	Tool description needs a history	Tool inspection returned "History unavailable" because no history identifier was supplied.
A2	Tool identifier not found	A tool identifier that does not resolve on the server (usually guessed or truncated).
A3	Conditional option structure	Nested conditional/repeat parameters submitted with a key structure the server rejects.
A4	Parameter value or data type	A value, option, format or collection type rejected by tool-state validation.
A5	Dataset or history identifier	Wrong, foreign, truncated or undecodable dataset/history identifiers, or histories not owned by the user.
A6	User-defined tool definition	User-defined tool representation rejected at creation (schema, pattern, discriminator or extra fields).
A7	Upload or file type	Unknown extension or failed upload/fetch.
A8	Server, connection or rate limit	HTML error pages, uncaught server exceptions, transport closure, time-outs, rate limits.
B1	User-defined tool: software missing	Package or executable absent from the container chosen for a user-defined tool.

Class	Name	Definition
B2	Job failed, no error message	A created job ended in error and no error message was returned to the agent.
B3	Job failed with error message	A created job failed while running and returned an error message or traceback.
B4	File format, compression or index	Input not parseable in the declared format, compression mismatch or missing index.
B5	Memory or compute limit	Memory exhaustion or killed process.
Z	Unclassified	Failure text matched no rule; retained for completeness.

Supplementary Note 4. Requested-versus-resolved parameter differences

The benchmark MCP layer compares the parameter values an agent requested with the tool state Galaxy resolved from them, and returns the difference as parameter provenance. Two statuses carry a difference: *validation_parameter_mismatch*, for which the harness blocked submission, and *parameter_mismatch*, for which the job executed. Within each, a call is an *explicit substitution* when a requested value was replaced (for example max → count) and a *missing path* when a requested parameter path is absent from the resolved state, typically because the value was placed under a conditional branch or repeat key that Galaxy did not select. Detected differences occurred in 4,352 tool-run calls: 3,436 blocked and 916 executed (Fig. 3e; Supplementary Table 37). Supplementary Data 3 lists the first substituted value of each call with an explicit substitution (2,085 calls). These are differences between request and resolved state; whether a given difference changed a scientific result was established only for the adjudicated cases in the failure ledger.

Supplementary Note 5. User-defined tool reliability measures

Every run `_galaxy_udt_and_wait` call returns a structured status (ok, failed, `udt_creation_failed` or none). For failed jobs, the failure_diagnostic block records the phase (before execution or during command rendering, versus command runtime) and whether stderr or a message was returned. A diagnostic-free failure was followed to the next tool-run call of the same run, which was classed as an identical resubmission when its tool inputs or UDT representation were byte-identical after key sorting, and otherwise as modified. Probe UDTs were identified by identifiers containing probe, preflight, render, smoke, diagnos(tic) or sanity. Of the calls that followed a diagnostic-free failure, 784 resubmitted an identical payload and 615 of those failed again (Extended Data Fig. 3a; Supplementary Table 28).

Supplementary Note 6. Failure adjudication protocol

Every rejected BixBench run (246: 135 open-ended code, 111 Galaxy) and every IWC run scoring below 0.5 (8) was read call by call. For each, we identified the decision point: the step at which the evidence the agent acted on diverged from what the reference required, and whether that evidence was sufficient. Rejected runs were contrasted with accepted runs of the same task, using the same configuration where one succeeded. Final answers were checked against independent evidence where possible (for example, reconstructing a statistic from the retained inputs, recomputing Benjamini–Hochberg adjustments, or comparing assembled contigs with the public *Agrius convolvuli* mitogenome OZ203683.1 by canonical 31-mers) rather than against the agent's own conclusion.

Each run received one primary and at most one secondary root-cause category. Galaxy is counted as *implicated* when a platform or tool defect is primary or secondary: 0 of 135 open-ended-code rejections and 22 of 111 Galaxy rejections.

Category	Definition	Open-ended code, primary / secondary (n = 135)	Galaxy, primary / secondary (n = 111)
Task under-specified or reference ambiguous	Task under-specified, or the reference depends on a choice the question does not state	64 / 26	58 / 12
Statistical or reasoning error	Statistical or reasoning error: weak evidence accepted, thresholds or invariants ignored, wrong denominator	39 / 63	26 / 52
Evaluator rejected a correct answer	Grader rejected an equivalent answer, or score records conflict	18 / 1	15 / 0
Missing domain knowledge	Domain-knowledge gap	12 / 6	3 / 3
Platform or tool defect	Galaxy, MCP or tooling defect (for code runs: local environment tooling)	0 / 0	4 / 18
No answer submitted	No answer written (turn ended or budget exhausted)	1 / 0	5 / 0
Output format violated	Output-format or identifier violation	1 / 0	0 / 0

Confidence was assigned per run: **high** (159), the decisive step is visible in the trace and confirmed by reproduction, independent reference comparison or contrast with an accepted sibling; **moderate** (55), the decisive step is identified but only partly verified; **mixed** (20), several contributing causes are present and the primary assignment is a

judgement; **unresolved** (12), no decisive step could be isolated and the category is the best-supported explanation. Mixed and unresolved assignments (32 of 246) are not proofs. The ledger (Supplementary Table 14) records the decision point for every run. For IWC, two Galaxy host-removal records scored 0.273 were re-examined: the submitted BWA-MEM output retained 20,899 read pairs, matching the BWA-route reference (20,896) rather than the Bowtie2-route reference (72,867) against which it was scored; the IWC archive itself performs no adjudication of these score-conflicted records (Extended Data Fig. 2d).

Supplementary Note 7. CompBioBench consensus proxy

CompBioBench archives 22 aggregate score vectors but no item-level grades or references, so task-level correctness cannot be observed. Answers were normalized (lower case, whitespace removed) and the modal answer across the 25 runs of each task was taken as the consensus. Applied to the 22 vectors with retained answers, the number of answers matching the consensus reproduces the archived totals with a mean absolute error of 2.6 of 100 and a mean bias of +2.0 (Extended Data Fig. 1c). On the 82 tasks with at least 20 of 25 concordant runs, a deviating answer is treated as a *probable* failure; this yields 33 Galaxy and 40 open-ended-code deviations (GPT-6 Astra excluded). The 18 contested tasks are not used. Re-fitting individual answers to the totals was rejected because it over-fits (22 constraints, 100 unknowns). The proxy is labelled as such wherever it appears and supports no task-level accuracy claim.

Supplementary Note 8. Replicate variability: non-unanimous triplicates and divergence mechanisms

A triplicate (cell) is one task run three times with one configuration in one environment. A triplicate is *non-unanimous* (a "mixed cell" in the draft text) when 1 or 2 of 3 BixBench replicates were accepted, when 1 or 2 of 3 CompBioBench replicates match a strong consensus answer (82 tasks), or when the within-cell range of IWC agreement scores exceeds 0.05. For every rejected replicate in a non-unanimous BixBench triplicate, the trace was compared with the accepted sibling(s) to identify what separated them. Mechanisms: **platform trap**, an optional interface path taken by this replicate only produced a silent default, a different output semantics or an undispatched job; **package-version drift**, the replicate installed a different software version from its accepted sibling; **convention or definition**, a domain convention or variable definition applied differently; **self-implemented method**, a hand-written script (or, in Galaxy, a UDT) diverged from the library method; **final-step slip**, a sort, count, complement or unit error after an otherwise correct analysis; **no answer**; and **benchmark-source lookup**, the only accepted replicate retrieved benchmark source data. Mechanisms describe what differed between replicates and are distinct from root causes (Note 6). Replicates are not seed-matched, so variability means run-to-run divergence under the same prompt, model and harness.

Supplementary Note 9. Statistical analysis and limitations

Unless stated otherwise, intervals are exploratory 95% percentile cluster-bootstrap intervals with 20,000 resamples (seed 20260922; NumPy linear quantiles; statistic-specific SHA-256 streams). Resampling units are source capsules for BixBench (up to 33), tasks for CompBioBench (up to 100) and tasks for IWC; configurations and replicate bundles are retained within each cluster, and cross-benchmark draws are independent and stratified by benchmark. Intervals assume independent clusters, which is not established for shared biological inputs, and are pointwise rather than multiplicity-adjusted. No confirmatory significance, equivalence, non-inferiority or causal claim is made; an interval spanning zero (differences) or one (ratios) is inconclusive. Token ratios divide the median input tokens of the three Galaxy runs by the median of the three code runs for the same task and configuration (all six totals required; cached input included; output and reasoning tokens not added), and are summarized by their median. Spearman correlations between prompt word count or recorded workload and failure markers are across tasks, use the three shared GPT configurations (nine observed runs required per task and environment) and 5,000 capsule or task bootstrap resamples with ties reranked within each resample.

Limitations. Benchmarks differ in task selection, prompts, execution budgets, exposed interfaces and endpoints, so cross-benchmark contrasts are descriptive. The IWC estimates rest on nine or ten task clusters. The superseded Claude Code harness ran BixBench only. Galaxy histories were snapshotted after the runs. Adjudication was performed by trace review, and 32 of 246 assignments are mixed or unresolved. Candidate recovery episodes were not adjudicated, and the code condition has no equivalent job-level detector. Shell exits are an operational marker, not a scientific failure rate. Human review time, reconstruction accuracy and monetary cost were not measured.

Supplementary Table legends

All tables are provided in Supplementary_Tables.xlsx, one sheet per table, with an index sheet mapping each table to its archive identifier and first citation.

Supplementary Table 1 | What can be combined fairly? The historical CompBio source composes final vectors from multiple campaigns, unlike a prospectively fixed independent replicate design. IWC has unbalanced 6 h and 12 h budgets and environment-specific prompts. Prompt additions, harnesses, tools, model verification, domain mix and campaign selection remain confounders. No cross-benchmark pooled accuracy denominator is constructed. Archive table X1.

Supplementary Table 2 | Original-evaluator acceptance and paired environment difference. Original binary evaluator scores are retained, including six scored missing answers (five Galaxy, one open-ended code). No regrading was performed. The superseded Claude Code configuration contributes 16 of the 24 extra accepted Galaxy runs; excluding it leaves eight. The original 100,000-resample audit interval was -1.41 to +7.91 percentage points; the fresh calculation above uses the common settings below. Intervals as in Supplementary Note 9. Archive table B1.

Supplementary Table 3 | Task coverage versus within-configuration advantage. Task coverage pools five configurations and three replicates (15 runs/task/environment), whereas the last column compares three-replicate counts within each of 250 matched task/configuration pairs. There are 198 tied pairs, and two tasks with no accepted answer in either environment. These endpoints do not describe the reliability of a single run. Archive table B10.

Supplementary Table 4 | Tasks with evaluator acceptance in only one condition. Exact rule: at least one accepted replicate across the five configurations in one condition and zero across all 15 runs in the other. Galaxy-only: zero tasks. Open-ended-code-only: one task, bix-45-q1. This is a task-level set comparison, not a test of superiority. Relative composition variability is the alignment statistic requested in the task. Model-specific counts are shown next. Archive table B11.

Supplementary Table 5 | Source-reported answer-vector scores, with provenance status. O = archive-labelled official, not an independently verified leaderboard receipt; P = predicted. 15 vectors are O and 10 P; 13 advertised hashes match retained bytes, nine mismatch and three vectors are absent. All parsed answers match for the 22 available vectors. Sol Galaxy r1/r3 were previously 92/100 P; dated metadata changes them to 93/100 O and 91/100 O. Neither version is silently discarded. No pooled accuracy, uncertainty interval, or task-level all/some/none-correct inference is computed from this mixture. Source: aggregate audit, score_vectors and score_conflicts. Archive table C1.

Supplementary Table 6 | CompBioBench cross-run consensus proxy and its validation against archived aggregate scores. CompBioBench retains aggregate score vectors but no item-level grades. For each task the modal normalized answer across its 25 runs was taken as the consensus; tasks with at least 20 of 25 concordant runs are "strong consensus". The proxy is validated by counting, for each archived vector with retained answers, how many of its 100 answers match the consensus (a). Deviations are probable, not adjudicated, failures. Source: CompBio/compBio_overview_audit.json and archived answers.

Supplementary Table 7 | Matched output agreement on the common nine IWC tasks. Agreement is the saved iwc-final-answer-v2.2 reference_accuracy. Its definition and tolerance are task-specific (Supplementary Table 9), so a mean is a task-weighted summary, not biological accuracy. Every column uses the same nine tasks: 27 runs per environment/configuration, 108 per environment in All four. Host removal is excluded from both environments for all models because of null and conflicted route scores (Supplementary Table 13). This is a post hoc sensitivity population, not a prespecified primary endpoint; all-ten-task summaries remain in Supplementary Table 9 and Supplementary Table 55. Three unsupported-route scores remain unknown, never zero. Differences average replicates within task, then weight tasks equally. For All four, it averages the 36 task/configuration differences. Intervals resample the nine tasks, 20,000 times, seed 20260922. The IWC report's 10,000-resample intervals (seed 20260923) differ slightly; the point estimates are identical and checked by the generator. With nine clusters, percentile coverage is approximate. No equivalence margin was prespecified. Archive table I1.

Supplementary Table 8 | Continuous replicate dispersion on the same nine tasks. One cell is one task, configuration and environment with three numeric replicates; all columns use the Supplementary Table 7 population. Within-cell range is maximum minus minimum across the three scores, summarized over nine cells. These continuous summaries introduce no pass threshold. Small dispersion can accompany consistently low agreement and is not correctness. With only three runs per cell, this is descriptive repeatability, not a stable variance estimate. Host-removal exclusions are retained in Supplementary Table 13; replicate labels are not matched seeds. Archive table I2.

Supplementary Table 9 | Task-level agreement, zero-scored runs and Galaxy job errors. All ten tasks are shown because IWC has too few tasks for a selection rule. Each environment has 12 runs per task (four configurations x three replicates). Minimums expose low-scoring runs that means can conceal; no distributional modality is assumed. Host removal has different model composition (12 versus nine numeric runs) and is not a matched contrast. Jobs are deduplicated within task and include legacy upload. Error jobs coexist with near-ceiling agreement: short-read QC has 22 Galaxy error jobs yet a minimum Galaxy agreement of 0.996. Score conflicts are records whose evaluator and run-record scores differ by more than 1e-9, or where one is null. Definitions come from retained evaluator metric records and are abbreviated here. Archive table I3.

Supplementary Table 10 | Does the same path instrument reproduce its agreement across benchmarks? Descriptive population: three shared GPT labels; all three run fingerprints observed and nonempty. This holds the extraction rule and included supplied labels fixed; eligible configuration proportions can still differ. IWC's higher Galaxy tool-set agreement fits workflow-defined tool chains, but rests on ten tasks. It does not control biological task complexity, command-vocabulary coverage, custom-wrapper identity or campaign selection. Lower agreement describes more variable recorded toolsets/command indicators, not worse science. Instrument values must not be directly compared across the two rows. Archive table X3.

Supplementary Table 11 | Task origin, recorded helper exposure and detected tool use. All benchmark-specific paired configurations. Requests, linkage and success use the Supplementary Table 24 rules. BixBench and CompBio exposure lists were not archived; observed requests show that the helper was available in at least those runs. Request rates and listed-run linkage counts have different denominators. A linkage count divided by all listed runs is a detected lower bound, not an estimate that treats missing transcripts/histories as zero use. IWC has no named-helper detections and its recorded lists omit that helper; neither observation establishes absence of alternative custom-code

interfaces. Tool Shed jobs outside the utility codebook (Supplementary Table 27) form a larger observed job share in IWC, but this classifier does not prove native completion of every task. Task origin and helper exposure are not experimentally separated. Local shell and interactive computation remain possible; Galaxy-only execution is uncertified in every benchmark. Archive table X19.

Supplementary Table 12 | How much of the environment difference rests on a few runs? Four decimals are kept because several differences are about $1e^{-3}$. Sign stable means all nine leave-one-task-out estimates share the full estimate's sign. The zero-run column drops every task with a zero-scored run for that configuration in either environment. It is an outcome-selected influence diagnostic, not a mechanism test or an improved performance estimate; it never replaces the full estimate. The last column adds host removal wherever all six runs are numeric. This drops the unmatched GPT-5.5 cell, and Luna then includes the two conflicted Galaxy records (Supplementary Table 13). The All four available-pair row weights 39 task/configuration pairs equally, so host removal has three configurations and other tasks four; it is not equal task weighting over ten tasks. The two larger positive differences shrink when tasks containing zero-scored runs are removed. Across configurations, the remaining differences range from 0.0010 to 0.0124. Archive table I4.

Supplementary Table 13 | Score provenance that changes an IWC conclusion. This table plays the role of Supplementary Table 5: score lineage is part of the result. The conflict tolerance is an absolute difference above $1e^{-9}$, or numeric versus null. Source: IWC scientific audit, `runs[*].score`, `run_record_acc` and `score_conflict_gt_1e_9`. Archive table I6.

Supplementary Table 14 | Failure ledger: every rejected BixBench run and every IWC run scoring below 0.5. One row per run (246 BixBench rejections: 111 Galaxy, 135 open-ended code; 8 IWC runs below 0.5). Run labels give environment (G, Galaxy; C, open-ended code), configuration (DS-Codex, DeepSeek V4 Pro via Codex; DS-ClaudeCode, DeepSeek V4 Pro via the superseded Claude Code harness) and replicate. Causes: SPEC, task under-specified or reference depends on an unstated choice; RIGOR, statistical or reasoning error (weak evidence accepted, thresholds or invariants ignored, wrong denominator); EVALUATOR, grader rejected an equivalent answer or score records conflict; KNOWLEDGE, domain-knowledge gap; PLATFORM, Galaxy, MCP or tooling defect (for code runs, local tooling); HARNESS, no answer written; CONTRACT, output-format or identifier violation. Confidence levels are defined in Supplementary Note 6. Source: trace-by-trace adjudication (Supplementary Note 6).

Supplementary Table 15 | Model dependence of the exclusive open-ended-code success. Single-task description: all 15 Galaxy answers were rejected, although all 45 retained non-fetch Galaxy jobs had state ok. An environment-wide missing execution capability is therefore not established. No benchmark-wide interval is estimated from one task. Archive table B12.

Supplementary Table 16 | Why bix-45-q1 differs: a version contrast retained within one original run. These p-values are outputs of the biomedical task, not tests of an environment effect. The recorded checks use rounded per-alignment values and two-sided asymptotic/automatic Mann-Whitney tests with continuity correction. The agent attributed the change to version-dependent gap/ambiguous-residue handling; the archived numerical contrast supports a version explanation for this run. It does not prove the intended reference release, adjudicate biological validity, or explain every rejected run. Source: task evidence, `open_ended_code_codex_gpt_5_6_sol_r1`, source lines 24, 33 and 45; trace messages and hashes are in `results.deep.case_sources`. No calculation was replayed. The within-run version contrast narrows the possible explanation beyond a generic input-assembly difference. Archive table B13.

Supplementary Table 17 | Evaluator-mode composition and acceptance. Modes are observational and task/configuration-dependent; these are not randomized verifier comparisons. The source audit reports byte-identical answers with different scores under different modes. Rounded-numeric scoring occurs only in the two DeepSeek configurations. These evaluator effects limit interpreting rejection as a demonstrated analytical mistake. Archive table B8.

Supplementary Table 18 | Completion is not correctness: every run scoring below 0.5. All 240 harness records report complete; completion alone does not establish final-artifact validity. The <0.5 selection is an auditor screening rule for inspection, not a benchmark failure threshold. Five of the eight runs are open-ended code runs. The only zero-scored Galaxy run is mitogenome assembly GPT-5.5 r1, and the same configuration and replicate also scored zero in open-ended code. The other two Galaxy rows are score-conflicted host-removal records, not established scientific failures; final adjudicated values are unknown. Reasons summarize saved evaluator fields where present; casing is documented in the IWC report. No answer was repaired or regraded. Archive table I5.

Supplementary Table 19 | Biological domain, answer convergence and operational burden. Domains come from task metadata; these are not difficulty strata. All four paired configurations are included. Single-answer counts use 12 submitted answers per task/environment. Error denominators include only detailed Galaxy runs. Ratios require all six usage records; small domains are descriptive case groups, not evidence of domain superiority. Archive table C4.

Supplementary Table 20 | Workbench capabilities visibly exercised by agents. All Galaxy configurations with retrieved transcripts; nonexclusive run-level counts of named calls. A completed helper call does not by itself prove job success or scientific validity. History creation/copying is not universally measured by these named helpers; it is documented in selected traces and must not be inferred for every run from a linked history. These counts support feature use, not improved human readability. IWC user-defined-tool detections are zero and that helper is absent from recorded lists; this association does not establish why custom code was or was not used. Archive table X14.

Supplementary Table 21 | Most frequently recorded Galaxy operations. Top six tool IDs by job count (ties sorted by full ID), among 5,042 deduplicated non-fetch jobs. Spreadsheet conversion, column selection and filtering accompany KEGG over-representation and PhyKIT metrics. KEGG is the Kyoto Encyclopedia of Genes and Genomes. Counts include preparation and inherited/later history state; operation frequency is not success attribution. Full IDs and event references are in the analysis JSON. Archive table B9.

Supplementary Table 22 | Most frequently recorded Galaxy operations. Top six tool IDs by job count among 16,686 deduplicated non-fetch jobs. AnnData inspection and bedtools intersection appear alongside filtering, column selection, Datamash and legacy upload. This is evidence of operation availability/use, not correctness, independent analyses or exclusive use of installed domain tools. Full IDs and event references are in the analysis JSON. Archive table C8.

Supplementary Table 23 | Most frequently recorded Galaxy operations. Top six tool IDs by job count among 1,352 deduplicated non-fetch jobs. Five of the six are domain workflow wrappers (read trimming, amplicon denoising, peptide verification, SRA metadata search), versus two of six in Supplementary Table 21 and Supplementary Table 22, where table utilities dominate. This fits tasks derived from Galaxy

workflows. The top operation is still a text utility (awk). Counts include inherited or later history state; frequency is not success attribution. Archive table I10.

Supplementary Table 24 | User-defined tools: explicit requests, tasks reached and linked execution. An explicit request is a `run_galaxy_udt_and_wait` event. Linked execution requires a declared `GalaxyUserTool` identifier matching a retained Galaxy job tool ID in the same run; success requires job state ok. Matching IDs supports linkage but does not establish exact chronology or authorship for inherited jobs. Requests alone do not prove submission or execution; unobserved requests are unknown. Four YAML-string representations are not parsed and remain linkage gaps. Jobs invoked through other interfaces can be missed; these are lower-bound detections, not an exhaustive custom-tool inventory. The proposed 30-40% claim depends on its denominator: task coverage, run use and confirmed execution are not interchangeable. IWC has zero detected named-helper requests and the helper is absent from its 120 recorded lists; alternative interfaces and unparsed/missing events are not excluded (Supplementary Table 11). Official user-defined tool documentation (<https://galaxyproject.org/tools/user-defined-tools/>) describes this capability and recommends existing published tools when suitable. Archive table X12.

Supplementary Table 25 | Are user-defined tool choices stable across repeat executions? One cell is one task/configuration; all three transcripts must be present. The three usage categories partition cells. Same prompt means identical archived prompt-file hashes; the stricter subset additionally requires a single verified runtime ID and reasoning setting. Variation is compatible with stochastic or context-dependent choice, not proof of randomness: input provenance, tool catalogs, prior state, campaign selection and services were not controlled. Missing runtime metadata is not treated as verified agreement. IWC runtime and reasoning come from invocation records; its all-zero cells describe this detector under different recorded helper lists, not a stable agent preference. Archive table X13.

Supplementary Table 26 | Why choose a user-defined tool? Distinguishing capability gaps from interface friction. Purposive mechanism cases, not percentages of all custom-tool use. `results.deep.case_sources` retains exact trace paths, hashes and line-numbered agent statements. The treeness counterexample establishes accepted output with retained PhyKIT jobs and no detected named UDT request, not an exhaustive native-only execution chain. To distinguish necessity from choice prospectively, snapshot catalog/search results, record rejected alternatives and parameter errors, then compare repeated matched tasks with a fixed tool catalog and an independently validated native workflow. The current archive cannot estimate the fraction used because no suitable Galaxy tool existed. Archive table X15.

Supplementary Table 27 | Observed helper exposure and Tool Shed use in IWC. No named `run_galaxy_udt_and_wait` event or request-linked UDT job was detected. That helper is absent from all 120 recorded `galaxy_mcp_tools` lists. A helper-list omission is evidence about the recorded interface, not proof that every API or shell route to custom code was unavailable or unused. The Tool Shed classifier excludes the declared `UTILITY_REPOS` text/table codebook; it is not an independent audit of domain-tool coverage or task completion. Each of 118 detailed histories contains at least one qualifying successful job, which does not demonstrate that all required operations used such jobs. Two pseudobulk histories include Jupyter interactive-tool jobs. Local shell analysis, arbitrary code within utilities, and inherited jobs are not excluded. Native-tool sufficiency, agent preference and strict Galaxy-only execution remain unestablished; two metadata-only histories are excluded from job denominators (Supplementary Table 11). Archive table I11.

Supplementary Table 28 | Reliability of the user-defined tool route. Counts are from Codex-harness Galaxy traces (1,908 runs), which return a structured status for every tool-run call. Failure phase and diagnostic availability come from the `failure_diagnostic` block of each failed job. A "no diagnostic" failure returns neither `stderr` nor a message. Probe UDTs are user-defined tools whose identifier names a probe, preflight, render, smoke, diagnostic or sanity check.

Supplementary Table 29 | Galaxy execution, missing evidence and recovery candidates. Jobs are deduplicated by server/native job ID and exclude data-fetch jobs, but include preparation and legacy upload. Histories may contain inherited/later state. A candidate is later same-tool/input success, not verified recovery of a scientific objective. 31/32 collection-limited runs had accepted answers. The original fresh-history check is false for all 750 Galaxy runs, despite retrieved history evidence; Galaxy-only completion remains unresolved. Archive table B4.

Supplementary Table 30 | Galaxy execution and recovery evidence. The archive contains 22,067 distinct creating jobs: 5,381 data-fetch and 16,686 non-fetch (including 372 upload1 jobs). Non-fetch jobs are not independent scientific attempts. Nonzero shell exits occur in 676/1,188 available Galaxy transcripts and 882/1,200 open-ended code transcripts; probes/searches can return nonzero, so these are not comparative scientific-failure rates. The two metadata-only histories contain 196 error-state datasets in `state_ids` but zero in `state_details`, plus 17 unaccounted elements. Those dataset states are not added to creating-job totals. Archive table C3.

Supplementary Table 31 | Galaxy execution, missing evidence and recovery candidates. Harmonized with Supplementary Table 29/Supplementary Table 30: jobs are deduplicated by server and native job ID, data fetch is excluded and legacy upload is kept. That gives 1,352 jobs, the IWC report's 1,330 non-acquisition jobs plus 22 upload1 jobs; error and deleted counts are identical. The two metadata-only histories are amplicon-denoising runs capped at 300 contents by the collector. Their failure counts are unknown, not zero. The 41 candidates collapse to 27 distinct successful endpoints and are not verified recoveries. No equivalent detector exists for open-ended code. Archive table I7.

Supplementary Table 32 | Specific recurring diagnostics and the mechanisms they suggest. Counts are deduplicated creating jobs; tool versions are pooled within the named family. The denominator contains every error job for that family, including missing/truncated diagnostics. Exact regex rules and matching job/source-event IDs are retained in the analysis JSON. These messages support input, datatype, identifier and design-validation hypotheses; they do not alone assign blame to the model, wrapper or platform. Multiple signatures may overlap. No claim is made that a tool always fails: the diagnostic probe with 57/57 error states in Supplementary Table 64 is not a biomedical-analysis success endpoint. Archive table X18.

Supplementary Table 33 | Which operational error messages are observable? All benchmark-specific paired configurations, not just shared GPT labels. Deduplicated non-fetch error jobs; case-insensitive regex matches on retained `stderr/stdout` excerpts, with overlapping categories. The exact codebook and every source event reference are in the analysis JSON. These are message indicators, not independently adjudicated root causes; missing/truncated logs suppress detection, and absent text does not mean no diagnostic existed on the server. No significance test compares these unequally observed taxonomies. Exit code zero does not override the recorded Galaxy error state. In IWC, `fastp` and `Bowtie2` error jobs retain no text at all (Supplementary Table 32), so the generic codebook under-detects its failures. Archive table X5.

Supplementary Table 34 | Classification of every failed Galaxy MCP call (A1–A8 before a job exists; B1–B5 after job creation). All Galaxy-environment runs with primary traces, both harnesses. Rules are applied in the listed order to the error text plus status of each failed call; the first match assigns the class. Rates use all Galaxy MCP calls in the benchmark as denominator. Example strings are verbatim, truncated and screened for credentials.

Supplementary Table 35 | Environment comparison using a shared, limited operational marker. Each pair requires six transcripts with at least one numeric shell exit code per run. The superseded Claude Code configuration is excluded because its exit codes are unobserved. Intervals resample capsules/tasks with all eligible configuration bundles. Only the shell channel is compared: searches, probes and package checks can return nonzero, while Galaxy job failures can be returned as successful API calls. Thus this marker does not answer whether Galaxy has more scientific failures or is intrinsically harder. Higher Galaxy input-token use (Supplementary Table 49/Supplementary Table 50), server-job errors (Supplementary Table 29/Supplementary Table 30) and concrete wrapper-binding friction (Supplementary Table 26) are distinct, supported forms of burden. Archive table X10.

Supplementary Table 36 | Adjudicated examples: correction versus changed objective. Purposive positive and contradictory examples, not a random sample of 213 candidates. Exact event IDs, parameter changes, timestamps and excerpts are preserved in case_reviews in the CompBio audit. No recovery success percentage is inferred from these two cases. Archive table C7.

Supplementary Table 37 | Requested-versus-resolved parameter differences in tool-run calls. The benchmark MCP layer compares the parameters an agent requested with the tool state Galaxy resolved and reports validation_parameter_mismatch (the harness blocked submission) or parameter_mismatch (the job executed). "Explicit value substitution" means a requested value was replaced; "requested path missing" means a requested parameter path is absent from the resolved state. Codex-harness traces only. Short tool identifiers are the Tool Shed repository names.

Supplementary Table 38 | Galaxy improvements suggested by the observed evidence. Priorities are analyst proposals, not measured intervention effects or claims that these features are wholly absent. Galaxy already documents history API and exports (https://docs.galaxyproject.org/en/release_26.1/_modules/galaxy/webapps/galaxy/api/histories.html), error troubleshooting (https://training.galaxyproject.org/training-material/faqs/galaxy/analysis_troubleshooting.html), datatype handling (https://training.galaxyproject.org/training-material/faqs/galaxy/datatypes_understanding_datatypes.html) and workflow reports (https://training.galaxyproject.org/training-material/faqs/galaxy/workflows_report_view.html). The proposed work is to assess and extend these capabilities at the agent interface. Collection limits belong to this audit's collector and are not established Galaxy server limits. Documentation checked 2026-09-23; no deployed-server feature audit was performed. Archive table X7.

Supplementary Table 39 | Common software families: installed Galaxy wrappers versus code-command indicators. Three shared GPT configurations. The last six families were added for IWC workflows; earlier rows are unchanged. Galaxy counts runs whose retained jobs name a Tool Shed wrapper from the family; open-ended code uses a declared command-name codebook, including PyDESeq2 as a DESeq2-family implementation. This inventory codebook is separate from the unchanged path-fingerprint vocabulary; shared family labels do not imply equivalent implementations or versions. Denominators require detailed histories or retrieved transcripts, respectively. Indicators are nonexclusive and their visibility differs: libraries inside custom tools can be hidden, and a command mention is not a verified invocation. This answers which families are observable in both conditions; it is not a fair count of equivalent scientific operations or proof of absent software. Workflow-derived task origin motivates checking tool coverage, but does not establish that every required tool/version was available on the recorded server. Archive table X9.

Supplementary Table 40 | Declared solution routes versus agreement. Method declarations were extracted on four tasks; 95/96 are available. Route counts use complete three-declaration cells, whereas means use all numeric runs in each row, including the incompletely declared cell. Host-removal means have unequal model composition. Routes are normalized declarations (for example Bowtie2 plus MACS2, DESeq2 or edgeR), not verified executions. Different declared routes coexist with near-ceiling agreement, which supports scoring valid alternatives against route-specific references. Route-specific references also mean agreement is conditional on route registration (Supplementary Table 13). Archive table I12.

Supplementary Table 41 | Observable triplicate path agreement under the harmonized rule. One cell = task x configuration x environment with three replicate labels. This report harmonizes BOTH benchmarks: all three fingerprints must be observed and nonempty. Identical / two identical / all distinct partition evaluable cells. Jaccard = intersection/union, averaged over the three replicate pairs and then over eligible cells. Galaxy uses version-stripped job tool IDs (excluding data fetch, retaining legacy upload); code uses the archived closed command vocabulary. These instruments cannot rank scientific consistency across environments; order, most parameters and biological validity are unmeasured. BixBench counts differ from its legacy report because that report also admitted partly missing/empty fingerprints. Archive table B5.

Supplementary Table 42 | Observable triplicate path agreement. One cell = task x configuration x environment with three replicate labels. This report harmonizes BOTH benchmarks: all three fingerprints must be observed and nonempty. Identical / two identical / all distinct partition evaluable cells. Jaccard = intersection/union, averaged over the three replicate pairs and then over eligible cells. Galaxy uses version-stripped job tool IDs (excluding data fetch, retaining legacy upload); code uses the archived closed command vocabulary. These instruments cannot rank scientific consistency across environments; order, most parameters and biological validity are unmeasured. BixBench counts differ from its legacy report because that report also admitted partly missing/empty fingerprints. Archive table C5.

Supplementary Table 43 | Observable triplicate path agreement under the harmonized rule. One cell = task x configuration x environment with three replicate labels. This report harmonizes all three benchmarks: all three fingerprints must be observed and nonempty. Identical / two identical / all distinct partition evaluable cells. Jaccard = intersection/union, averaged over the three replicate pairs and then over eligible cells. Galaxy uses version-stripped job tool IDs (excluding data fetch, retaining legacy upload); code uses the archived closed command vocabulary. These instruments cannot rank scientific consistency across environments; order, most parameters and biological validity are unmeasured. The legacy IWC fingerprint output is not used. Declared biological routes are in Supplementary Table 40. Archive table I8.

Supplementary Table 44 | Non-unanimous triplicates by benchmark, environment and configuration. A triplicate is one task run three times with one configuration in one environment (called a "mixed cell" in the draft text when non-unanimous). CompBioBench triplicates use the 82 strong-consensus tasks. IWC cells are variable when the range of the three agreement scores exceeds 0.05. The superseded Claude Code configuration was run on BixBench only.

Supplementary Table 45 | Task specification size and recorded workload versus operational failure markers. Three shared GPT configurations; nine observed runs are required per task/environment. Prompt word count is a pre-execution task-description-size proxy, not a validated difficulty score. Median job count is a post-execution workload measure; retries increase it, creating reverse causation. The job response divides recorded errors by recorded jobs to reduce the automatic opportunity effect of simply counting failures. Intervals use 5,000 capsule/task bootstrap resamples, seed 20260922, with ties reranked within each resample. These exploratory, unadjusted associations cannot establish that task complexity causes errors. Independent difficulty labels, budgets and attempt-level chronology are needed. Archive table X11.

Supplementary Table 46 | Different recorded paths, identical submitted answers: does the observation hold? All benchmark-specific paired configurations; three observed nonempty fingerprints are required. Answer identity uses exact archived submitted text after outer whitespace removal; it does not merge numerically close strings. Path variation means tool-ID/command-token sets differ, not independently adjudicated biological algorithms. BixBench can confirm acceptance; CompBio cannot confirm validity. The counts support multiple observed execution routes to the same answer, with these measurement limits. IWC is excluded: its final chat message is not the scored artifact; declared routes versus continuous agreement are in Supplementary Table 40. Archive table X16.

Supplementary Table 47 | Does a consistent recorded path imply acceptance? Descriptive association conditional on observable nonempty paths. Cells share tasks; failure-prone task composition can drive the pattern. Agreement is neither correctness nor causal evidence that varied methods help. The retained rows allow inspection of both accepted and rejected consistent paths. Archive table B6.

Supplementary Table 48 | What separated each rejected replicate from its accepted siblings in non-unanimous BixBench-Verified-50 triplicates. Every rejected replicate in a non-unanimous BixBench-Verified-50 triplicate (45 open-ended-code replicates in 33 triplicates; 36 Galaxy replicates in 28 triplicates) was compared call by call with the accepted replicate(s) of the same cell. The mechanism names what differed between the replicates, not why the answer was wrong (for that, see the failure ledger). Definitions in Supplementary Note 8.

Supplementary Table 49 | Input-token burden by configuration. Ratio = median input tokens of three Galaxy runs / median of three code runs for the same task and configuration; all six totals must be available. Q1-Q3 are quartile endpoints. Absolute medians use every available run in that row and are not the numerator/denominator of the paired median ratio. Cached input is already included. Output/reasoning are not added. Usage covers archived primary turns, not full campaign, compute or monetary cost. Failed runs are retained; missing usage is never zero. Archive table B7.

Supplementary Table 50 | Input-token burden by configuration. Ratio = median input tokens of three Galaxy runs / median of three code runs for the same task and configuration; all six totals must be available. Q1-Q3 are quartile endpoints. Absolute medians use every available run in that row and are not the numerator/denominator of the paired median ratio. Cached input is already included. Output/reasoning are not added. Usage covers archived primary turns, not full campaign, compute or monetary cost. Failed runs are retained; missing usage is never zero. Fourteen of 400 pairs are excluded (3 GPT-5.5, 10 Luna, 1 DeepSeek). The runtime-and-reasoning-verified sensitivity contains only 99 DeepSeek pairs (median 3.53), so it changes the configuration population. Astra usage is available in 100/100 unpaired open-ended code runs (median input 314,244.5); no Galaxy ratio is possible. Archive table C6.

Supplementary Table 51 | Input-token burden by configuration. Ratio = median input tokens of three Galaxy runs / median of three code runs for the same task and configuration; all six totals must be available. Q1-Q3 are quartile endpoints. Absolute medians use every available run in that row and are not the numerator/denominator of the paired median ratio. Cached input is already included. Output/reasoning are not added. Usage covers archived primary turns, not full campaign, compute or monetary cost. Failed runs are retained; missing usage is never zero. IWC usage is the audited terminal record, one per run, 240/240. Typical usage is higher in Galaxy, but summed input is lower: 815.8 million in Galaxy versus 919.6 million in open-ended code, because of a long open-ended code upper tail. The ratio medians reproduce the IWC report's 1.66-2.77 within-task ratios. With ten task clusters, every IWC interval includes one, so the typical direction is uncertain across the resampled task population. These intervals do not estimate monetary or causal overhead. Archive table I9.

Supplementary Table 52 | Does relative input-token overhead generalize across benchmarks? The contrast columns divide two benchmark-specific medians of within-task ratios; they are not absolute-token ratios. A value above one means greater relative overhead than in BixBench. IWC has ten task clusters, so its intervals are wide. Its lower point estimate cannot be attributed to installed wrappers: prompts, budgets, task selection and open-ended code tails also differ (Supplementary Table 51). Benchmark strata are resampled separately, preserving tasks/configurations within clusters. Selection differs and these are observational comparisons. Intervals as in Supplementary Note 9. Archive table X2.

Supplementary Table 53 | Control-plane burden and use of BioBlend or raw REST outside the MCP interface. Codex-harness Galaxy traces. Tool search and inspection = `search_galaxy_tools` and `inspect_galaxy_tool` calls. Returned characters are the lengths of the tool results returned to the agent, used as a proxy for context consumed because token totals cannot be attributed to individual calls. Inspected-but-never-executed counts distinct tool identifiers per run that were inspected but not submitted through `run_galaxy_tool_and_wait`, summed over runs. Shell commands are counted as Galaxy API calls when they use BioBlend or GalaxyInstance or call the Galaxy histories, datasets, jobs or tools API; bypass operations are matched among commands referencing BioBlend, GalaxyInstance or any `/api/` path. The seed history is supplied only in BixBench.

Supplementary Table 54 | Model trade-off: acceptance difference and token use relative to GPT-5.5. Reference is GPT-5.5 within the same environment and the same 50 tasks. Accuracy differences average all three replicates; token ratios divide task medians of three replicates and are then summarized by the median ratio. Error-run differences use the subset of tasks with all six detailed Galaxy runs and compare proportions with at least one error; the error column can therefore have a different task population from the other columns. Intervals use 20,000 paired capsule bootstrap resamples, seed 20260922. They are exploratory and pointwise. A difference interval containing zero does not establish equal performance or non-inferiority. Within Galaxy, Sol and Luna consumed more input tokens than GPT-5.5 without a resolved acceptance advantage; both DeepSeek configurations consumed more and had lower observed acceptance. These are selected-corpus associations, not general model rankings. Absolute pooled medians in Supplementary Table 58 and median paired ratios here can order models differently because they are different estimands. Archive table B15.

Supplementary Table 55 | All-task agreement, replicate dispersion, operational markers and tokens. IWC counterpart to Supplementary Table 58 and Supplementary Table 61, using continuous agreement instead of acceptance. Luna ran at max reasoning and the others at high, so configuration and reasoning are confounded. Budgets were unbalanced: 6 h or 12 h ceilings differ by configuration and

environment. Galaxy job errors and nonzero shell exits are different instruments, as in Supplementary Table 58. These are all-ten-task available-case summaries, not the matched population in Supplementary Table 7: GPT-5.5 code has nine scored tasks, other rows ten. Do not subtract unequal-population means. These are selected-corpus associations, not general model rankings or measured monetary costs. Archive table I13.

Supplementary Table 56 | Repeatability of accepted answers. Each row contains 50 task cells; the middle three categories are mutually exclusive. At least one is an overlapping endpoint, not single-run accuracy. These are replicate-labelled archived runs, not best-of-three adaptive attempts. Archive table B2.

Supplementary Table 57 | Largest acceptance discordances and tasks rejected throughout. Selection rule: absolute acceptance difference $\geq 5/15$ or zero acceptance in both environments; all five qualifying tasks are shown. Task-linked jobs are deduplicated within task. Operationally successful jobs do not establish a correct answer: RCV and transition/transversion results were rejected despite no recorded job errors. RCV means relative composition variability. The source reports implicate threshold choices, upstream gene sets and RCV input construction; those mechanisms are not independently adjudicated here. Archive table B3.

Supplementary Table 58 | Configuration outcomes, repeatability, operational markers and tokens. Each task contributes three runs/configuration/environment. Accepted-answer reliability is used for BixBench; CompBio can only report answer-text consistency, never correctness. Galaxy job errors and nonzero shell exits are different instruments. A shell exit may be a probe or failed search; Galaxy analysis is often offloaded to server jobs, so fewer shell exits do not establish easier execution. Shell denominators require at least one numeric recorded exit code: the superseded Claude Code traces contain no numeric exit codes and are unavailable, not zero failures. Tokens describe archived primary turns, not full execution cost. Missing data are excluded with explicit denominators; all observed outcomes are included. Archive table B14.

Supplementary Table 59 | Configuration-specific exclusive task successes hidden by pooling models. Same rule as Supplementary Table 4, but now applied within a task and configuration: one or more accepted replicates versus none. All 13 qualifying pairs are shown: nine Galaxy-exclusive pairs and four open-ended-code-exclusive pairs out of 250. These do not contradict Supplementary Table 4 because another configuration can succeed in the other environment. A one-of-three result is weak repeatability, not a stable platform advantage. Cells share tasks and cannot be treated as independent observations in a simple binomial test. Rows without an audited mechanism remain unresolved rather than being attributed to missing tools or model knowledge. Archive table B16.

Supplementary Table 60 | Usage completeness and repeated-answer consistency. Usage denominators are 300 runs/row; answer counts partition 100 triplicate task cells/row. Answers use archived submitted text with outer whitespace removed, not raw-file hashes. Text identity does not establish semantic equivalence or correctness. Among all 12 answers/task/environment, 62/100 Galaxy tasks and 58/100 open-ended code tasks have one distinct answer; answer sets differ across environments in 43/100 tasks. Archive table C2.

Supplementary Table 61 | Configuration outcomes, repeatability, operational markers and tokens. Each task contributes three runs/configuration/environment. Accepted-answer reliability is used for BixBench; CompBio can only report answer-text consistency, never correctness. Galaxy job errors and nonzero shell exits are different instruments. A shell exit may be a probe or failed search; Galaxy analysis is often offloaded to server jobs, so fewer shell exits do not establish easier execution. Shell denominators require at least one numeric recorded exit code: the superseded Claude Code traces contain no numeric exit codes and are unavailable, not zero failures. Tokens describe archived primary turns, not full execution cost. Missing data are excluded with explicit denominators; all observed outcomes are included. Archive table C9.

Supplementary Table 62 | Outlier case: what can actually be said about biomedical knowledge? The task asks which of ten tagAlign files comes from a different assay. Missing item scores prevent saying that most agents were wrong or assigning a biomedical-knowledge error rate. In GPT-5.5 Galaxy replicate 1, the trace records an unsuccessful external checksum lookup, then binned coverage correlations, chromosome distribution, strand correlation and transcription-start-site enrichment; it explicitly considers cell-line copy-number differences as an alternative explanation. These are documented diagnostic choices, not proof that the final index 9 is correct. Source: task evidence; trace lines 29, 30, 38 and 44 in results.deep.case_sources. Operational errors and uncertain scientific interpretation must remain separate. Archive table C10.

Supplementary Table 63 | Run-level operational burden with the shared configuration mix. Three shared GPT labels only; missing detailed histories are excluded explicitly. Each run contributes one binary indicator, preventing runs with many jobs from dominating this estimate. Histories may contain inherited/late state; this is recorded burden, not a causal benchmark/platform failure probability. No code-condition counterpart uses an equivalent job instrument. Archive table X4.

Supplementary Table 64 | Where recorded error jobs concentrate. Top five tool IDs ranked by error-job count within each benchmark (ties ordered by full ID); all non-fetch jobs and all configurations. These are triage targets, not tool-quality rankings: exposure, parameters, input quality, custom wrappers and task mix differ. In particular, udt-render-probe-v1 identifies a diagnostic probe; its error states cannot be assumed to be failed biomedical analyses. IWC error concentrations sit in domain workflow wrappers, the tools its tasks were built around. Full tool IDs and creating-job references are retained in jobs in the analysis JSON. Archive table X6.

Supplementary Table 65 | Claim strength, remaining questions and evidence needed. All 160 task evidence paths, hashes, included run IDs and original finding IDs are linked in the source manifest. Table-specific display values and calculated statistics, run/cell denominators, job references and exclusion rules are in the analysis JSON. Source task evidence is unchanged; no hidden references, recovered agent code, new Galaxy runs or new correctness judgements were used. Archive table X8.

Supplementary Table 66 | Testing the six motivating observations. These verdicts evaluate the motivating statements, not a prespecified hypothesis set. All statistical intervals are exploratory. Model capacity, workbench functionality, operational friction, scientific correctness and execution cost remain separate endpoints. Archive table X17.

Supplementary Table 67 | Distinct benchmark endpoints in the shared-label descriptive set. Three shared supplied GPT labels, not verified identical runtime/reasoning settings across benchmarks. Endpoints differ, so read rows side by side, never pooled or ranked. BixBench differences are percentage points of acceptance; IWC differences are agreement units on 0-1. All IWC columns use the same nine-task subset as Supplementary Table 7; host-removal nulls and conflicts are documented in Supplementary Table 13, not relabelled as

failures. A BixBench zero denotes evaluator rejection; an IWC zero can reflect a contract check or output disagreement and is not an interchangeable scientific-failure endpoint. CompBio answer-text identity is omitted from this outcome table because it is neither a ceiling score nor correctness (see Supplementary Table 60). Task sets, prompts, budgets, interfaces and endpoints differ; whether workflow-derived task origin narrows an environment effect is not identifiable from this comparison. Archive table X20.

Supplementary Data legends

Supplementary Data 1 | Per-run summary of all 4,240 archived runs. One row per run with benchmark, task, configuration, environment, replicate, archived score, submitted answer, evaluator mode, token usage, shell and MCP call counts and failed-call counts by class. Reference answers are omitted to limit benchmark contamination.

Supplementary Data 2 | All 7,389 failed Galaxy MCP calls. One row per failed call with run, trace line, MCP tool, status, failure class, verbatim error text and arguments (truncated to 1,500 characters; screened for credentials).

Supplementary Data 3 | 2,085 detected parameter substitutions. For each Codex-harness tool-run call with an explicit requested-versus-resolved substitution, the first substituted parameter path with requested and resolved values.